

Report of Calibration
PSC ALSu_1ST_ARTICLE_4-CH-SLOW_MSSS/N 0001
2026-09-25 15:05:35

Calibration Current standard: BNL PSC ATE S/N 001
Calibration Volt standard: HP 3458A-002 S/N 2823A 23647
Calibration Resistance standard: Fluke 742A-1 S/N 1063008
End Header

lab{2}Chan1
Burden resistor = 33.3333

Measuring initial gains and offsets

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.004365	1.003152	-1.013167	-1.014435	1.003391	0.027590
-27.001745	26.998786	-26.809563	-26.833082	27.039328	0.029584

	dacSP	dcct1	dcct2	dacRB
Initial measured offsets:	0.001146	-0.016566	-0.016974	0.000000
Initial measured gains:	0.999933	0.992269	0.993125	1.001550
Gain corrections:	0.999933	1.007791	1.006923	1.000000

Writing gain and offset corrections for dacSP, dcct1, and dcct2 to PSC

Measuring DAC readback gain and offset

DAC SP	DAC RB
1.000000	0.999114
27.000000	27.037889

Measured offset: -0.002377
Measured gain: 1.001491
Gain correction: 0.998511

Writing gain and offset constants for dacRB to PSC

Verification

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.004401	1.003944	-1.004749	-1.004869	1.004103	-0.040428
-27.001529	27.001040	-27.001446	-27.001434	27.000196	-0.001182

	dacSP	dcct1	dcct2	dacRB
Final measured offsets:	0.000456	-0.000364	-0.000490	0.000198
Final measured gains:	0.999999	0.999983	0.999978	0.999961

	dacSP	dcct1	dcct2	dacRB
Final measured offsets mean:	0.000228	-0.000256	-0.000432	0.000068
Final measured offsets stdev:	0.000155	0.000129	0.000157	0.000099
Final measured gains mean:	0.999996	0.999995	0.999987	0.999984
Final measured gains stdev:	0.000017	0.000015	0.000012	0.000018

Saving channel 1 calibration constants to qspi

lab{2}Chan2
Burden resistor = 33.3333

Measuring initial gains and offsets

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.003930	1.006973	-1.015497	-1.013209	1.003456	-0.050595
-27.001544	26.999940	-26.805576	-26.812265	27.073624	0.009780

	dacSP	dcct1	dcct2	dacRB
Initial measured offsets:	-0.003223	-0.019581	-0.016947	0.000000
Initial measured gains:	0.999821	0.992017	0.992362	1.002970
Gain corrections:	0.999821	1.008047	1.007696	1.000000

Writing gain and offset corrections for dacSP, dcct1, and dcct2 to PSC

Measuring DAC readback gain and offset

DAC SP	DAC RB
1.000000	0.999690
27.000000	27.072050

Measured offset: -0.003093
Measured gain: 1.002783
Gain correction: 0.997225

Writing gain and offset constants for dacRB to PSC

Verification

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.003953	1.003965	-1.004035	-1.003857	1.003692	0.004355
-27.001397	27.001332	-27.001339	-27.001110	27.001078	0.034830

	dacSP	dcct1	dcct2	dacRB
Final measured offsets:	-0.000015	-0.000087	0.000089	-0.000274
Final measured gains:	0.999997	0.999995	0.999993	1.000001

	dacSP	dcct1	dcct2	dacRB
Final measured offsets mean:	0.000047	-0.000235	-0.000222	-0.000205
Final measured offsets stdev:	0.000139	0.000179	0.000227	0.000184
Final measured gains mean:	0.999989	0.999988	0.999987	1.000016
Final measured gains stdev:	0.000005	0.000010	0.000010	0.000008

Saving channel 2 calibration constants to qspi

lab{2}Chan3
Burden resistor = 33.3333

Measuring initial gains and offsets

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.003657	1.006928	-1.013442	-1.014197	1.006296	0.020045
-27.001245	27.005804	-26.821690	-26.822750	27.050440	0.019327

	dacSP	dcct1	dcct2	dacRB
Initial measured offsets:	-0.003221	-0.017095	-0.017838	0.000000
Initial measured gains:	1.000050	0.992717	0.992729	1.001741
Gain corrections:	1.000050	1.007336	1.007325	1.000000

Writing gain and offset corrections for dacSP, dcct1, and dcct2 to PSC

Measuring DAC readback gain and offset

DAC SP	DAC RB
1.000000	1.002968
27.000000	27.049080

Measured offset: 0.001195
Measured gain: 1.001774
Gain correction: 0.998230

Writing gain and offset constants for dacRB to PSC

Verification

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.003394	1.003539	-1.003729	-1.003819	1.004443	0.033726
-27.001163	27.000985	-27.000946	-27.001379	27.001549	0.043161

	dacSP	dcct1	dcct2	dacRB
Final measured offsets:	-0.000158	-0.000357	-0.000434	0.000917
Final measured gains:	0.999988	0.999979	0.999992	0.999987

	dacSP	dcct1	dcct2	dacRB
Final measured offsets mean:	0.000133	-0.000194	-0.000203	0.000996
Final measured offsets stdev:	0.000273	0.000184	0.000173	0.000284
Final measured gains mean:	0.999993	0.999991	0.999993	0.999991
Final measured gains stdev:	0.000007	0.000014	0.000007	0.000009

Saving channel 3 calibration constants to qspi

lab{2}Chan4
Burden resistor = 33.3333

Measuring initial gains and offsets

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.003285	1.004470	-1.012472	-1.012121	1.003276	0.031874
-27.001205	27.006443	-26.820053	-26.804596	27.092779	0.029664

	dacSP	dcct1	dcct2	dacRB
Initial measured offsets:	-0.001029	-0.016532	-0.016764	0.000000
Initial measured gains:	1.000156	0.992679	0.992098	1.003366
Gain corrections:	1.000156	1.007375	1.007965	1.000000

Writing gain and offset corrections for dacSP, dcct1, and dcct2 to PSC

Measuring DAC readback gain and offset

DAC SP DAC RB

1.000000 0.999756
27.000000 27.090963

Measured offset: -0.003752

Measured gain: 1.003508

Gain correction: 0.996504

Writing gain and offset constants for dacRB to PSC

Verification

Itest	dacSP	dcct1	dcct2	dacRB	err
-1.003563	1.003310	-1.003807	-1.003617	1.004202	0.001869
-27.001269	27.000603	-27.001118	-27.001278	27.001446	-0.003070

	dacSP	dcct1	dcct2	dacRB
Final measured offsets:	0.000237	-0.000260	-0.000055	0.000894
Final measured gains:	0.999984	0.999985	0.999998	0.999998

	dacSP	dcct1	dcct2	dacRB
Final measured offsets mean:	0.000094	-0.000142	-0.000090	0.000777
Final measured offsets stdev:	0.000197	0.000131	0.000043	0.000129
Final measured gains mean:	1.000003	0.999997	1.000002	0.999993
Final measured gains stdev:	0.000012	0.000007	0.000005	0.000017

Saving channel 4 calibration constants to qspi

Test data reviewed by _____ Date_____